Benchao Su

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Academic Interests

My research interest lies in the p-adic local Langlands program, especially in locally analytic representations coming from p-adic geometry.

Employment

IMJ-PRG, Sorbonne Université

- Postdoctoral Researcher (2026.09–present).

Education

Peking University

- Ph.D. in Pure Mathematics (2021–2026).
- Advisor: Yiwen Ding.

Publications and Preprints

4. **Locally analytic vectors in the completed cohomology of quaternionic Shimura curves** (joint with Zhenghui Li, Zhixiang Wu). arXiv:2601.13625.
3. **De Rham cohomology of Lubin-Tate spaces and Drinfeld spaces**. arXiv:2503.09352.
2. **On the locally analytic Ext^1-conjecture in the $\mathrm{GL}_2(L)$ case**. arXiv:2504.17683.
1. **Locally analytic vectors in the completed cohomology of unitary Shimura curves** (joint with Qiu Tian). arXiv:2505.10290.

Talks

9. *On the Ext algebra of some supercuspidal locally analytic representations of* $\mathrm{GL}_2(\mathbb{Q}_p)$, Sino-French Number Theory Workshop, Tianyuan Mathematical International Exchange Center, 2026.07, Yunnan.
8. *Locally analytic vectors in the completed cohomology of quaternionic Shimura curves*, Thematic Month, Around the Langlands program, p-adic aspects of the Langlands program, 2026.2, Marseille.
7. *On the Lie algebra cohomology of de Rham complex of Drinfeld tower*, Seminar at Claude Bernard Lyon 1 University and ENS de Lyon, 2025.11, Lyon.
6. *On admissibility of de Rham complex of Drinfeld tower of dimension* 1 *over* L, Seminar at University Clermont Auvergne, 2025.11, Clermont-Ferrand.
5. *Solid locally analytic representations and solid almost quasi-coherent sheaves*, Recent progress in p-adic geometry, 2025.03, Yunnan.
4. *On Breuil's locally analytic* Ext^1*-conjecture in the* $\mathrm{GL}_2(L)$ *case*, ENTR Workshop 24, 2024.09, Essen.
3. *Translations and the locally analytic* Ext^1 *conjecture in the* $\mathrm{GL}_2(L)$ *case*, p-adic geometry and analysis, 2024.06, Shenzhen.
2. *Translations and the locally analytic* Ext^1 *conjecture in the* $\mathrm{GL}_2(L)$ *case*, p-adic Day, 2024.05, Beijing.
1. *Locally σ-analytic vectors in the completed cohomology of unitary Shimura curves*, Mini-conference on p-adic Langlands Program, 2023.08, Beijing.

[1] Updated October 27, 2026